THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS · MEDIUM LEVEL UNIT TEST SET 2

Chapters 5–8: **Circles, Perimeter and Area, Probability, Sequences and Progressions** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. The sum of the first 13 even natural numbers is: 1

(a) 169 **(b)** 182 **(c)** 195 **(d)** 208

2. $\angle AOB = 96^\circ$ at the centre and C lies on the major arc AB. Then $\angle ACB =$ 1

(a) 38° **(b)** 48° **(c)** 58° **(d)** 68°

3. The area of a quadrilateral whose diagonal is 20 cm and whose offsets from the diagonal are 6 cm and 8 cm is (cm^2): 1

(a) 142 **(b)** 140 **(c)** 141 **(d)** 139

4. A letter is chosen at random from the word MISSISSIPPI. The probability that it is the letter S is: 1

(a) $\frac{1}{3}$ **(b)** $\frac{5}{11}$ **(c)** $\frac{4}{11}$ **(d)** $\frac{7}{11}$

5. The sum of the first 14 odd natural numbers is: 1

(a) 210 **(b)** 196 **(c)** 224 **(d)** 182

6. $\angle AOB = 90^\circ$ at the centre and D lies on the minor arc AB. Then $\angle ADB =$ 1

(a) 125° **(b)** 135° **(c)** 145° **(d)** 155°

7. A rectangle of perimeter 50 cm has length 18 cm. Its area (cm^2) is: 1

(a) 127 **(b)** 128 **(c)** 126 **(d)** 125

8. In a town of 12000 people the probability that a randomly chosen person owns a smartphone is $\frac{3}{5}$. The expected number of owners is: 1

(a) 7202 **(b)** 7200 **(c)** 7199 **(d)** 7201

9. The ratio of the 5th term to the 3rd term of a GP is 9. The common ratio (positive) is: 1

(a) 3 **(b)** 2 **(c)** 5 **(d)** 4

10. In a circle, a chord of length 18 cm is at distance 12 cm from the centre. Another chord of the same length is at a distance (in cm): 1

(a) 11 **(b)** 13 **(c)** 14 **(d)** 12

11. A circular wire of radius 7 cm (take $\pi = \frac{22}{7}$) is bent into a square. The side of the square (cm) is: 1

(a) 11 **(b)** 12 **(c)** 13 **(d)** 10

12. If $P(E) = 5/12$, then the probability that E does not happen is: 1

(a) $\frac{7}{12}$ **(b)** $\frac{7}{13}$ **(c)** $\frac{2}{3}$ **(d)** $\frac{5}{12}$

13. The number of multiples of 7 between 20 and 150 (both excluded) is: 1

(a) 18 **(b)** 19 **(c)** 21 **(d)** 20

14. AB is a diameter and C lies on the circle with $\angle ABC = 24^\circ$. Then $\angle BAC =$ 1

(a) 86° **(b)** 76° **(c)** 66° **(d)** 56°

15. A wheel of diameter 70 cm (take $\pi = \frac{22}{7}$) makes 50 revolutions. The distance covered (in metres) is: 1

(a) 110 **(b)** 111 **(c)** 109 **(d)** 112

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. Find the area of a quadrilateral $ABCD$ in which diagonal $AC = 26$ cm and the perpendiculars from B and D to AC are 9 cm and 7 cm. 2

17. A bag has 6 red and 9 blue balls. 3 more red balls are added. Find the new probability of drawing a red ball. 2

18. Find the sum of all multiples of 5 between 10 and 101. 2

19. In a circle, $\angle AOB = 68^\circ$ with O the centre. Find $\angle OAB$ and $\angle ACB$ (C on the major arc). 2

20. Find the arc length for 40° in a circle of radius 63 cm (take $\pi = \frac{22}{7}$). 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. The 3rd term of an AP is 16 and the 8th term is 41. Find the AP and its 20th term. 3

22. O is the centre of a circle and $\angle AOB = 72^\circ$. A point C lies on the major arc and D on the minor arc. Find $\angle OAB$, $\angle ACB$ and $\angle ADB$. 3

23. The sides of a parallelogram are 15 cm and 12 cm and the height on the side 15 cm is 8 cm. Find its area and the height on the side 12 cm. 3

24. A bag has 3 red and 9 blue balls. How many red balls must be added so that the probability of drawing a red ball becomes $\frac{1}{2}$? 3

OR

Find the sum of the first 20 terms of the AP whose first term is 5 and whose 8th term is 26. 3

25. Two parallel chords of lengths 14 cm and 48 cm lie on the same side of the centre of a circle of radius 25 cm. Find the distance between them. 3

OR

Find the area of the cyclic quadrilateral with sides 8 cm, 9 cm, 12 cm and 19 cm. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

26. Two chords of the same circle have lengths 10 cm and 24 cm and the circle has radius 13 cm. Find the distances of the chords from the centre and say which chord is nearer to the centre. 5

OR

A triangular field has sides 13 m, 13 m and 10 m. Find its area, and the cost of ploughing it at ₹5 per m^2. 5

27. A factory tested 200 bulbs and found 8 defective. Find the probability that a bulb is defective, and estimate the number of defective bulbs in a lot of 5000. 5

OR

A sum of ₹10000 is invested at 10% per year compound interest. Show that the amounts at the end of years form a GP and find the amount after 3 years. 5

END OF QUESTION PAPER **Total: 50 Marks**

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